Why real Magic

Run drc check; find width violation on thin metal1

The idea

- GOLDEN: drc_error_count=1; thin on metal1

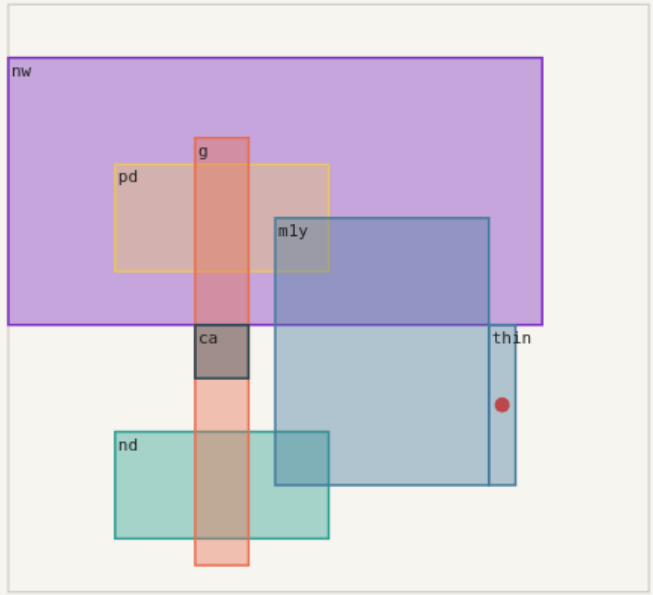
drc check

REAL DRC

drc check

Step 1 / 5 · cmd · module07-03-real-drc

cmd



Explanation

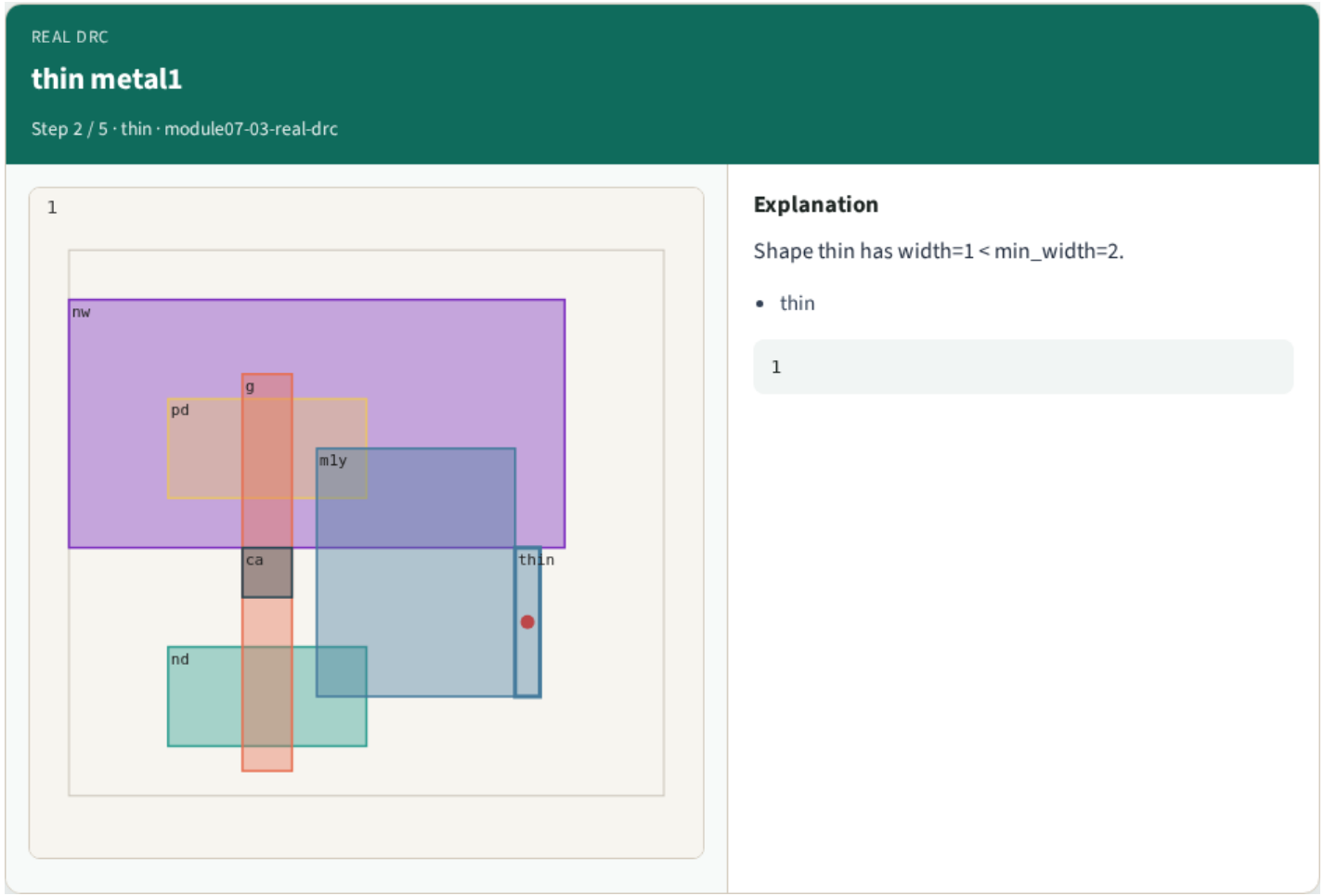
drc check scans planes against tech rules.

- drc_check

```
cmd
```

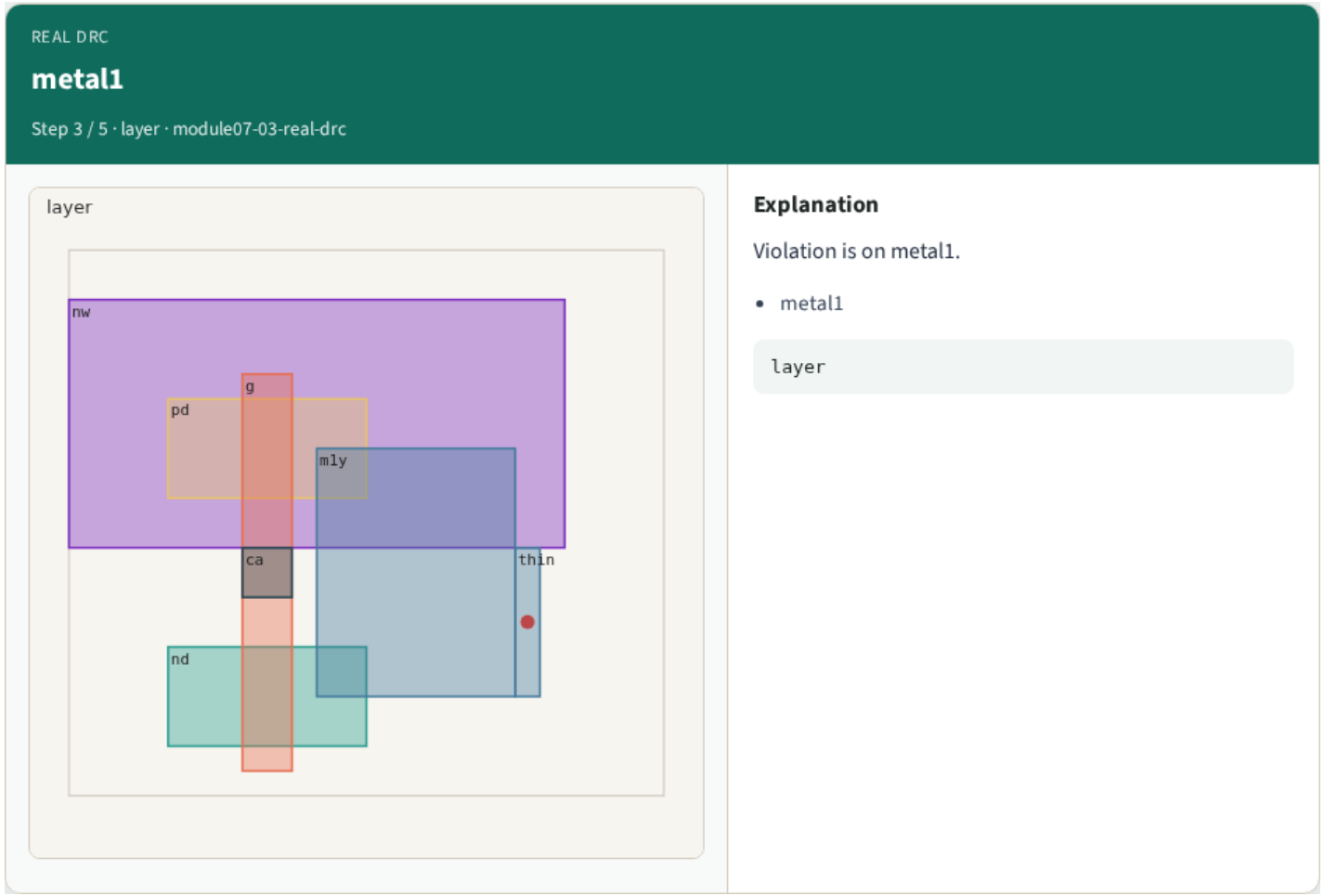
drc check

thin metal1



thin metal1

metal1



metal1

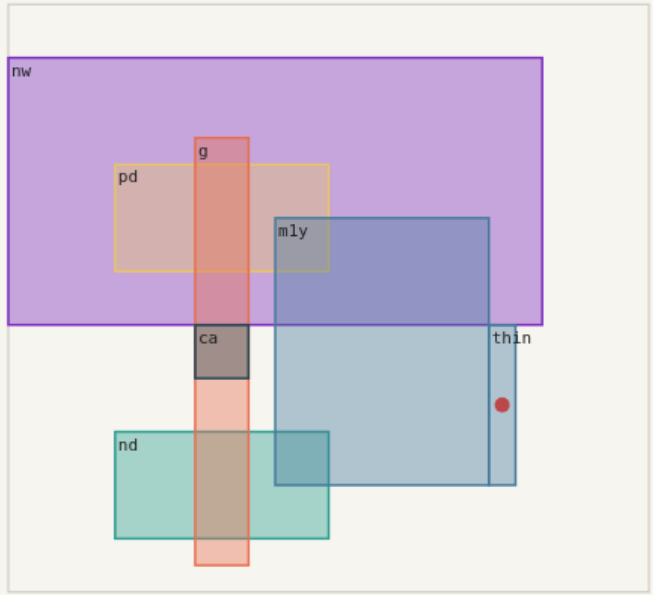
One error

REAL DRC

One error

Step 4 / 5 · count · module07-03-real-drc

1



Explanation

Teaching log reports DRC_ERRORS=1.

- DRC

```
1
```

One error

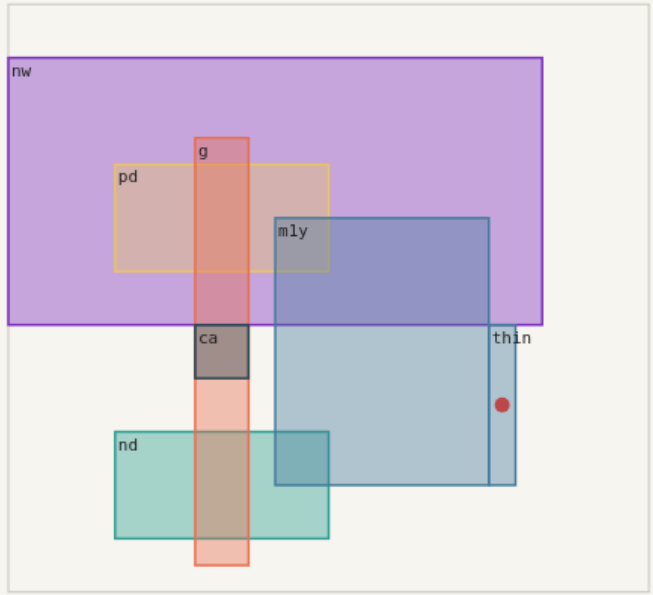
Golden

REAL DRC

Golden

Step 5 / 5 · golden · module07-03-real-drc

1



Explanation

GOLDEN: drc_error_count=1 on thin.

- GOLDEN

```
1
```

Golden

Browser lab track

- Open **real-drc**
- Build the Magic recipe or open evidence stages, then Check

Implement track

- From common/, run real_solvers.py and confirm REAL_MAGIC_OK

Pitfalls

- Editing goldens by hand without re-running scripts
- Confusing Part A map JSON with checked-in logs

Try these

- From the common folder, run the real Magic self-test

Try these — try these

```
cd courses/learn_magic/common
python3 real_solvers.py
```

Pseudocode

- Here is the check in plain steps

Pseudocode — try these

```
drc_error_count=1; thin on metal1.
ok if goldens match
```

Next

- Complete the quiz for this part
- Continue the Part B evidence track, then algorithms and internals